tf.einsum Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/einsum

> *Tensor contraction over specified indices and outer product.*

Function Signatures

```
tf.einsum( equation, *inputs, **kwargs )
```

Code Examples

```
tf.einsum(
equation, *inputs, **kwargs
)
```


```
tf.einsum(
equation, *inputs, **kwargs
)
```


```
C[i,k] = sum_j A[i,j] * B[j,k]
```


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C[i,k] = sum_j A[i,j] * B[j,k]
```


```
m0 = tf.random.normal(shape=[2, 3])
m1 = tf.random.normal(shape=[3, 5])
e = tf.einsum('ij,jk->ik', m0, m1)
# output[i,k] = sum_j m0[i,j] * m1[j, k]
print(e.shape)
(2, 5)
```

```
m0 = tf.random.normal(shape=[2, 3])
```

Documentation

Tensor contraction over specified indices and outer product.

View aliases

Main aliases

tf.linalg.einsum

See
Migration guide for
more details.

tf.compat.v1.einsum, tf.compat.v1.linalg.einsum

Used in the notebooks

- Matrix approximation with Core APIs

- Advanced automatic differentiation
- Working with RNNs
- Neural style transfer
- Quantum data
- Modeling COVID-19 spread in Europe and the effect of interventions
- Parametrized Quantum Circuits for Reinforcement Learning
- Research tools

Einsum allows defining Tensors by defining their element-wise computation. This computation is defined by equation, a shorthand form based on Einstein summation. As an example, consider multiplying two matrices A and B to form a matrix C. The elements of C are given by:

$$C_{i,k} = \sum_j A_{i,j} B_{j,k}$$

The corresponding einsum equation is:

In general, to convert the element-wise equation into the equation string, use the following procedure (intermediate strings for matrix multiplication example provided in parentheses):

- remove variable names, brackets, and commas, (ik = sum_j ij * jk)
- replace "*" with ",", (ik = sum_j ij , jk)
- drop summation signs, and (ik = ij, jk)
- move the output to the right, while replacing "=" with "->". (ij,jk->ik)

Many common operations can be expressed in this way. For example:

Matrix multiplication

Repeated indices are summed if the output indices are not specified.

Dot product

Outer product

Transpose

Batch matrix multiplication

This method does not support broadcasting on named-axes. All axes with matching labels should have the same length. If you have length-1 axes, use tf.squeeze or tf.reshape to eliminate them.

To write code that is agnostic to the number of indices in the input use an ellipsis. The ellipsis is a placeholder for "whatever other indices fit here".

For example, to perform a NumPy-style broadcasting-batch-matrix multiplication where the matrix multiply acts on the last two axes of the input, use:

Einsum will broadcast over axes covered by the ellipsis.

- optimize: Optimization strategy to use to find contraction path using

opt_einsum. Must be 'greedy', 'optimal', 'branch-2', 'branch-all' or 'auto'. (optional, default: 'greedy').

- name: A name for the operation (optional).

Returns

Raises

- the format of equation is incorrect,
- number of inputs or their shapes are inconsistent with equation.